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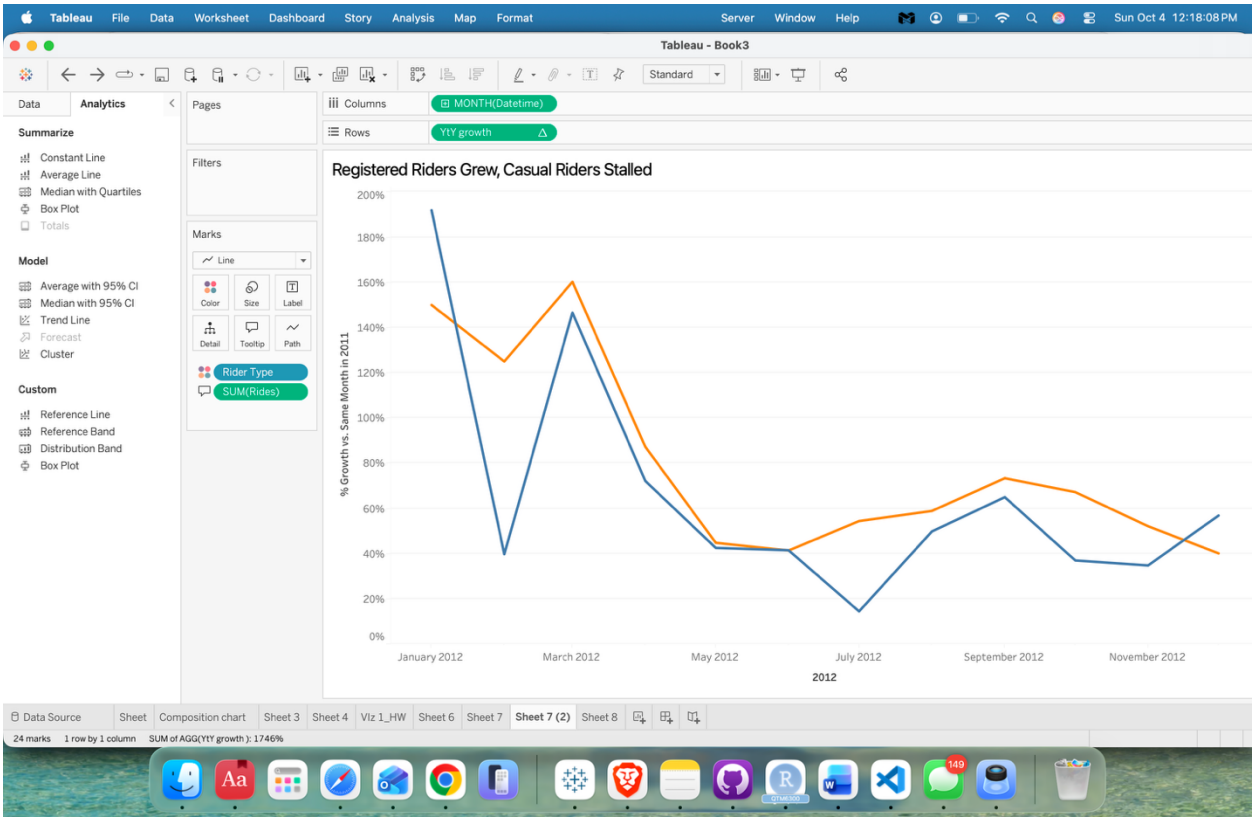
Professor Karst

Case: Bike Share

Viz 1: Business Question

Is ridership growing year over year, and are casual riders growing as fast as registered riders?

	Datetime	
Rider Type	2011	2012
Casual	247,252	372,765
Registered	995,851	1,676,811



I used Claude to help write a calculated field that compares each month of 2012 with the same month in 2011, since Tableau's built-in Year Over Year option wasn't available for this chart.

```
(SUM([Rides]) - LOOKUP(SUM([Rides]), -12)) / LOOKUP(SUM([Rides]), -12)
```

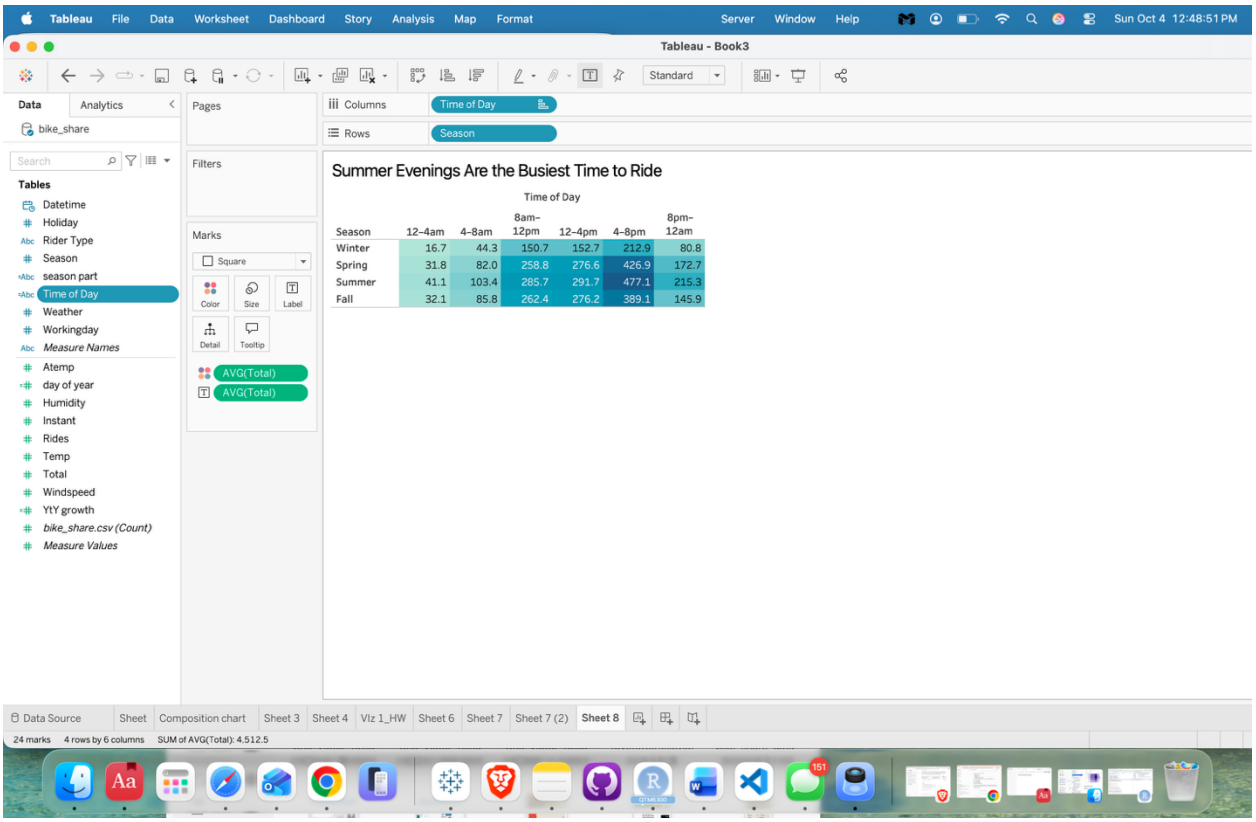
Viz 2: Writeup

This chart shows the total number of rides each month from January 2011 to December 2012. The orange line represents registered members and the blue line represents casual riders, meaning people without a membership. Both lines follow the same seasonal pattern: ridership rises in spring, peaks in summer, and drops in winter. Both groups also grew from 2011 to 2012, with total rides rising about 65%. However, the two groups grew in different ways. Registered riders kept climbing through summer 2012 and reached a high of about 175,000 rides in September, roughly 68% more than the year before. Casual riders grew more slowly, at about 51%. After a sharp peak of about 36,000 rides in July 2011, casual ridership rose quickly in spring 2012 but then stayed flat at around 42,000–44,000 rides per month from May through September, while registered riders continued to grow.

This flat stretch suggests casual growth hit a limit in summer 2012, which is normally the best season for leisure riding. Leadership should find out what drove the July 2011 spike and why it didn't happen again. If a promotion or local event caused it, repeating a similar summer campaign could restart casual growth. If bikes were running out at busy stations because of the jump in registered riders, adding bikes or rebalancing more often during summer weekends would give casual riders a better chance to ride. Since casual riders are the main source of future members, getting their growth back on track matters for long-term growth.

Viz 2: Business Question

When is demand for bikes highest, and how should we make sure bikes are available at those times?



I used Claude to help write the calculated field that groups each hour into four-hour time blocks for this heatmap. Tableau doesn't have a built-in way to group hours into custom blocks, so this formula assigns each ride to a block based on the hour it started.

```
IF DATEPART('hour', [Datetime]) < 4 THEN "12–4am"
ELSEIF DATEPART('hour', [Datetime]) < 8 THEN "4–8am"
ELSEIF DATEPART('hour', [Datetime]) < 12 THEN "8am–12pm"
ELSEIF DATEPART('hour', [Datetime]) < 16 THEN "12–4pm"
ELSEIF DATEPART('hour', [Datetime]) < 20 THEN "4–8pm"
ELSE "8pm–12am"
END
```

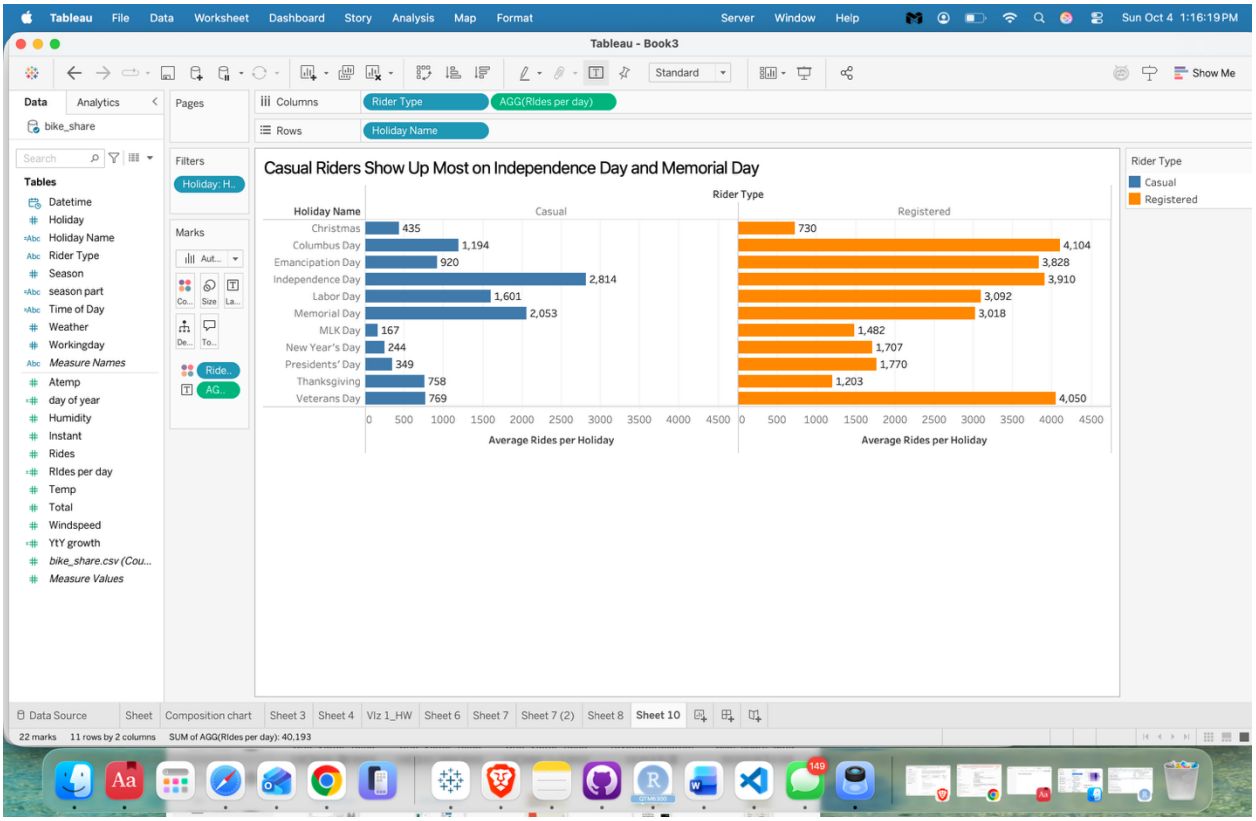
Viz 2: Writeup

This heatmap shows the average number of rides per hour for all riders, broken down by season (rows) and time of day in four-hour blocks (columns). Darker cells mean more rides. Demand is lowest overnight, between 12 and 4am, with fewer than 45 rides per hour in every season. It rises through the day and peaks between 4 and 8pm, when people are commuting home and riding for leisure. The busiest time of the year is summer evenings, which average about 477 rides per hour. That's more than double the same time block in winter (213) and over 11 times the overnight level. Spring and fall evenings are close behind, at about 427 and 389 rides per hour.

This connects directly to the finding in Visualization 1 that casual rider growth stalled in summer 2012. Summer afternoons and evenings are when demand is highest and bikes are most likely to run out. They are also the hours when casual riders, who mostly ride for leisure, want to ride. If registered commuters are taking most of the available bikes during these peak hours, casual riders may be turned away, which could explain why their growth slowed. To prevent this, the company should rebalance bikes to busy stations before the 4pm rush in spring and summer and consider adding bikes at high-demand stations during peak season. The overnight hours, when demand is lowest, are the best time to move bikes into position for the next day.

Viz 3: Business Question

Which holidays bring out the most riders, and when should the company target casual riders?



The data only marks whether a day is a holiday, so Claude helped write a field naming each holiday by date, plus a field averaging rides per holiday, since New Year's Day appears only once in the data:

```
IF MONTH([Datetime]) = 1 AND DAY([Datetime]) <= 2 THEN "New Year's Day"
ELSEIF MONTH([Datetime]) = 1 THEN "MLK Day"
ELSEIF MONTH([Datetime]) = 2 THEN "Presidents' Day"
ELSEIF MONTH([Datetime]) = 4 THEN "Emancipation Day"
ELSEIF MONTH([Datetime]) = 5 THEN "Memorial Day"
ELSEIF MONTH([Datetime]) = 7 THEN "Independence Day"
ELSEIF MONTH([Datetime]) = 9 THEN "Labor Day"
```

```
ELSEIF MONTH([Datetime]) = 10 THEN "Columbus Day"
ELSEIF MONTH([Datetime]) = 11 AND DAY([Datetime]) < 15 THEN "Veterans Day"
ELSEIF MONTH([Datetime]) = 11 THEN "Thanksgiving"
ELSEIF MONTH([Datetime]) = 12 THEN "Christmas"
END
```

Viz 3: Writeup

This chart shows the average number of rides on each holiday in 2011 and 2012, with casual riders (blue) on the left and registered members (orange) on the right. Both panels use the same scale, so bar lengths can be compared directly. The two groups behave very differently on holidays. Registered members ride most on Columbus Day, Veterans Day, and Independence Day, with around 4,000 rides each. Many businesses stay open on Columbus Day and Veterans Day, so members likely keep commuting as usual. Their ridership drops sharply on holidays when nearly everyone has the day off, such as Christmas (730) and Thanksgiving (1,203). Casual riders, by contrast, show up most on summer holidays. Independence Day averages about 2,814 casual rides and Memorial Day about 2,053, reaching about 70% of the registered level, even though casual riders account for only about 19% of all rides over the two years.

These summer holidays are the company's best opportunity to reach casual riders. The surge likely includes visitors and occasional riders coming out for holiday events, such as the July 4th celebrations on the National Mall. This connects to Visualizations 1 and 2: casual growth slowed in summer 2012, and summer afternoons and evenings are when bikes are in highest demand. To capture this opportunity, the company should run promotions around Independence Day and Memorial Day, such as discounted day passes and membership sign-up offers for casual riders. It should also make sure extra bikes are stocked near event areas so casual riders aren't turned away. Converting even a small share of these holiday riders into members would support long-term growth.